

How the daily rate sheet gets made

Pricing engines, LLPAs, MSR values, pipeline hedging, best execution and lock desk policy - the full loop, from overnight market data to the price a borrower sees.

The one-sentence version: every morning a pricing engine combines capital-markets data, loan-level risk adjustments, servicing values and the lender's own margins into a rate sheet; through the day, hedge marks trigger reprices, and lock desk policies defend the book - with every measured cost recycled into tomorrow's sheet.

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1. The pricing engine and the daily rate sheet

A product and pricing engine (PPE) - Optimal Blue, Polly, Lender Price, or a lender's internal system such as UWM's - is the machine that turns raw market data into the rate/price grid loan officers and brokers see each morning. Overnight and pre-market, the capital markets desk reads the forward market where conventional loans are effectively sold (TBA trading in UMBS), and that sets a raw base price for each note rate. The engine then layers on risk-based adjustments, stretches pricing across lock periods, and bakes in the lender's margin and compensation structure - including lender-paid compensation (LPC) for brokers.

Four categories of inputs

1. Capital markets data

- MBS / TBA prices (UMBS coupons)
- 10-yr Treasury yields
- Agency guarantee fees (g-fees)
- Servicing values (SRP / MSR)
- Investor / correspondent bids

2. Loan-level adjustments (LLPAs)

- FICO score and LTV / CLTV
- Occupancy (primary, second, investment)
- Property type and units
- Loan purpose and loan amount
- Subordinate liens, state adjusters

3. Product and lock terms

- Program: conventional, FHA, VA, jumbo
- Term: 30 / 20 / 15 yr, fixed or ARM
- Lock period: 15 / 30 / 45 / 60 days
- Escrow waiver option
- Buydowns and discount points

4. Lender business inputs

- Target profit margin
- Compensation plans (LPC / BPC)
- Hedging and pipeline costs
- Capacity and volume goals
- Competitor rate positioning

What comes out

The published rate sheet contains a **base price grid** (price by note rate and lock period), **adjustment tables** (LLPAs plus any lender overlays), and **lock desk policies** (extension, relock and renegotiation rules). It typically publishes mid-morning Eastern once MBS trading gives a clear read, and reprices intraday whenever the market moves beyond the lender's tolerance - sometimes several times on volatile days.

2. Loan-level price adjustments (LLPAs)

LLPAs are risk-based pricing hits (or credits) applied per loan, quoted in points - percentages of the loan amount - and layered cumulatively on top of the base price. Fannie Mae and Freddie Mac publish the core grids; lenders may add their own overlays. Because they compound, the same note rate can carry very different prices for two different borrowers.

The main factors and their direction

Factor	Typical direction of impact
FICO x LTV (the core matrix)	Lower FICO and higher LTV each add cost; the combination compounds. A 640 FICO / 95% LTV loan can carry ~3 points more in LLPAs than a 780 / 60% loan.
Occupancy	Investment properties add roughly 2-4 points; second homes add meaningfully as well. Primary residences are the baseline.
Loan purpose	Cash-out refinances add ~0.4-2+ points over purchase/rate-term, scaling with FICO and LTV.
Property type	Condos above ~70% LTV, 2-4 unit properties, and manufactured homes each add cost.
Subordinate financing	A second lien behind the new first adds an adjustment based on CLTV.
Loan amount / balance	Very small balances and high-balance (conforming-jumbo) loans carry adjustments.
DTI / other	Certain programs and eras have carried DTI-related or product-specific adjusters; lender overlays vary.

How an LLPA becomes a rate

LLPAs are charged in price, not rate - but borrowers usually absorb them as rate. If a loan accumulates 1.5 points of LLPAs, the borrower can pay 1.5% of the loan amount in cash at closing, or take a higher note rate whose richer price absorbs the hit (roughly 0.25% in rate per 1.0-1.25 in price, varying with the market). This is why two neighbors with identical loan amounts can be quoted rates half a point apart on the same day from the same lender.

Connection to the rate sheet: the LLPA tables are printed on (or embedded in) the daily rate sheet as adjustment matrices. The pricing engine applies them automatically when a loan is priced, so the 'base' price is something almost no borrower actually receives.

3. Mortgage servicing rights (MSR) valuation

When a servicer or investor prices MSRs, the question is: *what is the present value of the future servicing fee income from these loans, minus the cost to service them, adjusted for how long the loans will actually last?* The servicing fee looks tiny - roughly a quarter point per year on conventional loans - but it lasts as long as the loan does, so expected loan life dominates the value.

Six categories of factors

1. Servicing cash flows

- Service fee strip: 25-44 bps / yr
- Escrow float income
- Ancillary and late fees
- Average loan balance
- Recapture / refi opportunity

2. Prepayment risk (the biggest driver)

- Note rate vs current market rate
- Expected CPR (prepayment speed)
- Seasoning and burnout
- Housing turnover by geography
- Expected life of the cash flows

3. Credit and default risk

- FICO, LTV and DTI mix
- Expected delinquency rates
- Delinquent servicing cost: 5-15x normal
- State foreclosure timelines (judicial vs not)
- Product: FHA / VA vs conventional

4. Cost to service

- Performing: ~\$90-120 per loan per year
- Non-performing: \$500+ per loan per year
- P&I; and T&I; advance obligations
- Compliance and technology
- Boarding / transfer costs

5. Pool characteristics

- Escrowed vs non-escrowed share
- Geographic concentration
- Loan size distribution
- Delinquency status at sale
- Rep and warranty recourse

6. Market conditions

- Discount rate / required yield
- Rate outlook and volatility
- Hedging costs
- MSR supply and demand
- Capital and liquidity rules

A loan with a note rate well above the current market is a refinance waiting to happen - its servicing may last only two or three years, so the MSR is worth little. A low-rate loan the borrower will never refinance can pay the fee for a decade. Credit quality works on the cost side: delinquent loans are actively expensive because the servicer funds collections and loss mitigation and (especially on Ginnie Mae / FHA / VA pools) advances missed principal, interest, taxes and insurance out of pocket while awaiting reimbursement.

Typical result: MSR prices are quoted as a multiple of the service fee or in basis points of unpaid balance - commonly around 4-5x the fee, i.e. roughly 1.0-1.3% of loan balance for performing conventional servicing. Actual levels move daily with rates and the MSR trade market.

4. The natural hedge and a sample MSR bid grid

The natural hedge

MSR values move opposite to the rest of the mortgage business. When rates **fall**, everyone refinances: origination desks print money, but the servicing book evaporates as low-fee annuities pay off early, so MSR values collapse. When rates **rise**, applications dry up, but nobody refinances out of the portfolio, so the servicing strip lasts years longer and MSR values swell. A lender holding both businesses rides a roughly flat combined line instead of either rollercoaster - which is why large originators such as UWM, Rocket and Pennymac retain servicing. Note the floor: prepayments never reach zero, because people still move, divorce and pass away, so there is a base 'housing turnover' speed of roughly 5-6 CPR no matter how out-of-the-money a refinance is.

Sample bid grid - conventional 30-year, bulk package

An MSR buyer quotes a bulk package roughly like this. Price falls as note rate rises, purely because higher-rate loans are more likely to refinance away. The multiple is price divided by service fee - '5x' means five years of fee income paid up front. (Illustrative levels.)

Note rate bucket	Service fee	Assumed CPR	Price (bps of UPB)	Multiple
≤ 5.50%	25 bps	6	135	5.4x
5.51 - 6.00%	25 bps	7	125	5.0x
6.01 - 6.50%	25 bps	9	114	4.6x
6.51 - 7.00%	25 bps	13	100	4.0x
> 7.00%	25 bps	19	85	3.4x

Common bid adjustments

Characteristic	Typical adjustment
Non-escrowed loans	-3 to -6 bps (no float income)
Average balance below ~\$150k	-8 to -12 bps (same per-loan cost, less fee income)
Judicial-foreclosure-state heavy	-2 to -5 bps (longer, costlier default timelines)
30+ day delinquent loans	Carved out of the bid or deeply haircut
Ginnie Mae (FHA / VA) collateral	Quoted on a separate, lower grid (advance obligations)
High escrow balances	Small premium (more float)

Full-circle connection: when a lender builds the morning base price, the expected MSR value (or the servicing-released premium a buyer will pay) is added on top of the MBS execution. Richer servicing values mean a more aggressive rate sheet within days - so FICO, loan size and escrows influence a borrower's rate twice: once through LLPAs, and again through what the servicing strip is worth.

5. Pipeline hedging with TBAs

A rate lock is a free option the lender hands the borrower. Between 'borrower locks' and 'loan funds' - typically 15 to 60 days - the lender is exposed: if rates rise before closing, the loan it promised to fund at yesterday's rate is worth less in the market. The hedge desk neutralizes this by selling TBA MBS forward (going short) against the pipeline.

Pull-through: hedging expected fundings, not the whole pipeline

Pipeline stage	Typical pull-through (share that will close)
Application stage	~70%
Underwritten / approved	~85%
Clear to close	~97%

Hedge amount = pipeline balance x pull-through probability, computed loan by loan and summed. Stage migrations alone change coverage needs: a loan moving from application (70% weight) to clear-to-close (97% weight) adds coverage need without any new lock.

The two scenarios

If rates rise

- Pipeline loans lose value
- Short TBA position gains
- Pull-through goes up (nobody re-shops)
- Net: roughly offset

If rates fall

- Pipeline loans gain value
- Short TBA position loses
- Fallout jumps - borrowers re-shop
- Net: hedge slippage (convexity cost)

The asymmetry in the right-hand box is **negative convexity**: if rates rise the borrower keeps the lock and the hedge protects the lender; if rates fall the borrower can walk and re-lock elsewhere, so the lender loses on the TBA short *and* loses the loan. The expected cost of that free option is a line item in the margin on the rate sheet, and lock desk policies (Section 8) exist specifically to blunt it.

6. Intraday rebalancing and repricing mechanics

Rebalancing the hedge

The desk runs a running scoreboard: expected fundings on one side, TBA shorts on the other. Hedge platforms (MCT, Optimal Blue Hedging, Polly) pull the pipeline from the loan origination system every 15-60 minutes, mark it against live TBA quotes, and flag when net exposure drifts outside a tolerance band - commonly plus or minus 1-2% of the pipeline. Inside the band, do nothing (trading costs money); outside it, trade back to neutral, usually in round \$1M lots. New locks add shorts; fundings and withdrawals retire them; stage migrations shift the weights.

Rate moves change both sides at once: a rally cuts pull-through (borrowers re-shop) so the desk buys TBAs back at now-higher prices; a selloff spikes pull-through so the desk sells more TBAs at now-lower prices. Systematically buying high and selling low is the realized cost of the borrower's free option.

Reprice mechanics

Element	How it typically works
Morning cushion	The sheet publishes (~10:00 ET) with a small buffer in the margin - on the order of an eighth of a point of price - to absorb normal noise.
Trigger	Current-coupon TBA moves beyond a threshold from the level used at publication - often 10-15 ticks, roughly 0.3-0.5 in price on volatile days.
Asymmetry	Lenders reprice for the worse fast (protects margin) and for the better slowly (discretionary, often only if competitors move).
Adverse selection	The moment the market sells off, LOs watching MBS feeds rush to lock at the stale price. Slow reprices get 'picked off' - panic locks land above market and the hedge covers them at lower levels.
Frequency	One reprice on a normal day is common; three or four in an afternoon around a hot CPI print or Fed day is not unusual.

The loop: the hedge desk's live mark on the pipeline is the same signal that trips the reprice; the reprice throttles how much mispriced volume can flow in; whatever slippage still occurs is measured and fed back into tomorrow's margin. The rate sheet and the hedge position are one book viewed from two sides.

7. Loan sale execution: cash window, securitization, aggregators

Once a loan funds on the warehouse line, the lender must turn it back into cash. The 'best execution' comparison runs daily, per loan, alongside the rate sheet build.

Execution path	What it is	Who uses it / trade-offs
GSE cash window	Deliver the whole loan to Fannie or Freddie for cash. No pooling logistics.	Smaller lenders; simple; the GSEs cap volume per seller (low billions per year each).
Securitization (swap)	Pool loans into UMBS and sell into the deep TBA market.	Scale players; usually a few extra ticks of price; pairs naturally with retaining servicing.
Aggregator sale	Sell the whole loan to a correspondent aggregator, who pays a servicing-released premium (SRP) on top.	Most small correspondents; effectively is the 'release' decision - servicing goes with the loan.

Retain vs release servicing

Retain servicing

- Keep the MSR asset (~1.0-1.3% of balance)
- Natural hedge for origination income
- Keep the customer and recapture opportunity
- Ties up capital; requires servicing ops

Release servicing

- Cash SRP today (~1% of balance)
- No servicing operation or advances
- Hand the buyer the relationship
- Give up recapture on the next refi

Large lenders run best-execution software that compares every path per loan using live cash window prices, TBA levels, aggregator bid tapes and their own MSR marks. The winner of that comparison flows straight back into the base price on tomorrow's rate sheet.

8. Lock desk policies: extensions, relocks, renegotiations

Automated reprices protect the *new* pipeline; lock desk policies protect the desk from the locks it already holds. Each policy prices or constrains a specific way borrowers (or loan officers) could exploit the free option in a rate lock.

Policy	When it applies	Typical terms	The economics
Extension	Closing is delayed past lock expiry	~1.5-2.5 bps per day; common blocks of 7 / 15 / 30 days	Mirrors the TBA roll cost of pushing the hedge to the next settlement month plus carry - the desk passes through its cost of waiting.
Relock	Lock lapsed with the loan unfunded	Worst-case pricing: worse of the original or current market, for 30-60 days	Kills the game of letting a lock expire during a rally for a free do-over.
Renegotiation (float-down)	Market rallied significantly after lock	Allowed once; trigger of roughly 25-50 bps improvement; borrower gets part of the gain	Deliberately gives back part of the option to avoid losing all of it - refusing entirely means the borrower walks and the hedge loss is realized anyway.

Pull-through scorecards

Wholesale lenders track every broker's and LO's pull-through, extension usage and renegotiation behavior. A shop that habitually locks at three lenders and delivers to one is harvesting free options from the other two; it eventually sees worse pricing, tighter lock terms, or loses access. The measured behavior is quantified in basis points and folded into the margin input of the pricing engine.

9. The full feedback loop and glossary

The loop, end to end

1. Overnight, capital markets reads TBA/UMBS prices, Treasuries, g-fees and servicing values, and sets base prices. **2.** The pricing engine layers on LLPAs, lock-period spreads, hedge/option costs and margin (including LPC), and publishes the morning rate sheet. **3.** Locks flow in; the hedge desk shorts TBAs against pull-through-weighted expected fundings and rebalances all day. **4.** If TBAs move beyond tolerance, a reprice goes out within minutes. **5.** Funded loans exit through the best execution among cash window, securitization and aggregator sale, with servicing retained or released. **6.** Extension, relock, renegotiation and fallout costs are measured, scorecards updated - and everything feeds tomorrow's margin. The borrower's rate quote is the visible tip of this entire machine.

Glossary

Term	Meaning
TBA	'To be announced' - the forward market for agency MBS where conforming loans are effectively sold; the benchmark for daily pricing.
UMBS	Uniform MBS - the single security for Fannie Mae and Freddie Mac conventional loans.
LLPA	Loan-level price adjustment - risk-based pricing add-on quoted in points of the loan amount.
MSR	Mortgage servicing right - the asset representing the right to collect the servicing fee strip.
SRP	Servicing released premium - cash paid for servicing when a loan is sold servicing-released.
CPR	Conditional prepayment rate - annualized share of a pool expected to prepay.
Pull-through	Share of locked loans that actually close and fund.
Fallout	Locked loans that never fund - the flip side of pull-through.
Negative convexity	The lender loses more when rates fall (fallout + hedge loss) than it gains when rates rise - the cost of the borrower's free lock option.
LPC / BPC	Lender-paid / borrower-paid compensation - how the broker is compensated on a wholesale loan.
G-fee	Guarantee fee charged by Fannie/Freddie for taking the credit risk of a loan in an MBS.
Best execution	Daily per-loan comparison of all sale paths to find the highest all-in price.
Warehouse line	Short-term credit facility that funds loans between closing and sale.
Current coupon	The TBA coupon trading nearest par - the market's reference for new production.

Levels quoted throughout (LLPA magnitudes, MSR multiples, extension fees, reprice thresholds) are typical ranges for illustration; actual values vary by lender, program and market conditions and move daily. This guide is educational, not financial advice.